

Observability Analysis and EKF-Based Sensorless Control of Permanent Magnet Synchronous Machine

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Abstract: The observability analysis is presented and a combining method applying extended Kalman filter (EKF) and high frequency injection (HFI) method is proposed to realize the estimation of rotor speed and position at standstill and in normal situation for permanent magnet synchronous machine (PMSM). Considering the observability of start-up situation, a surface-mounted permanent magnet synchronous machine (SPMSM) model is developed using vector control method, then the high frequency injection methods are carried out to estimate the rotor speed and position accurately at standstill, while the extended Kalman filter algorithm is applied to validate the observability. Simulation results are shown that the proposed method can estimate the rotor position and speed accurately.

Key Words: observability analysis, permanent magnet synchronous machine (PMSM), sensorless control, high frequency injection (HFI), extended Kalman filter (EKF)

1 Introduction

Permanent Magnet Synchronous Motor (PMSM) has a high torque to inertia ratio, high speed, high efficiency as well as high reliability and compact size. These qualities make PMSM becomes one of the most applicable AC machines and PMSM is therefore gaining extensive research attention in recent years [1, 2]. Since mechanic position sensor is usually too expensive, which increases the cost and decrease the stability of the system, mechanic sensorless control is becoming a research focus now.

When it comes to the sensorless techniques, many research works have been done in speed and position estimation of PMSM using state observer. Some of them use the back electromotive force (EMF) to determine the speed and position of the rotor [3], others with the adjunction of a Kalman filter [4] or sliding mode observer [5]. Whereas the main problem of the PMSM observer-based sensorless techniques is the deteriorate performance in low and zero speed operation conditions, the usual solution is to inject a superimposed signal to the control values in order to generate information without interfering with the normal functioning of motor [6, 7]. Those solutions show good performance at low speed and standstill but are obviously very dependent on geometrical or magnetic saliency of the rotor to allow extraction of the position and speed information through filtering. Thus they are unusable for surface-mounted PMSM.

In this paper, a new state observer based on extended Kalman filter and high frequency injection (HFI) method is used to observe the rotor position and speed. On the one hand, the injected high frequency signal makes sure the observability problem at zero speed. On the other hand, extended Kalman filter can solve nonlinear equation directly by numeric iteration. The proposed method combining EKF and HFI offers an effective mean to ensure the observability in unobservable region, thus solving the problems incurred in sensorless control of PMSM and observability analysis. The simulation results verify the proposed method can reach

the satisfying estimation results both in standstill situation and in normal situation.

2 Mathematical Model

In the stator reference frame, the PMSM dynamic mathematical model can be written as [8]:

$$\begin{aligned}\frac{d\mathcal{I}}{dt} &= \mathcal{L}^{-1}(\mathcal{V} - \mathcal{R}^{eq}\mathcal{I} - \psi_r \mathcal{C}'(\theta)\omega) \\ \frac{d\omega}{dt} &= \frac{p}{J}(T_m - T_l) \\ \frac{d\theta}{dt} &= \omega\end{aligned}\quad (1)$$

where $\mathcal{I} = [i_\alpha \ i_\beta]^T$ and $\mathcal{V} = [v_\alpha \ v_\beta]^T$ stand for currents and voltages vectors in the $\alpha - \beta$ reference frame, ω is the electrical speed of the rotor, θ is the electrical position, ψ_r is the rotor permanent magnet flux. $\mathcal{L}$ is the inductance matrix, $\mathcal{R}^{eq}$ is the equivalent resistance matrix:

$$\mathcal{R}^{eq} = \mathcal{R} + \omega \mathcal{L}' = \begin{bmatrix} R_s & 0 \\ 0 & R_s \end{bmatrix} + \omega \frac{\partial \mathcal{L}}{\partial \theta} \quad (2)$$

where R_s is the resistance of one stator winding. p is the number of pole pairs, J is the inertia of the rotor with the load, T_l is the resistant torque, and T_m is the motor torque. $\mathcal{C}'(\theta)$ denotes the partial derivative of $\mathcal{C}(\theta)$ with respect to θ :

$$\mathcal{C}(\theta) = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} \quad ; \quad \mathcal{C}'(\theta) = \frac{\partial \mathcal{C}(\theta)}{\partial \theta} = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix} \quad (3)$$

For a salient rotor PMSM, $\mathcal{L}$ represents the inductance matrix and is depend on the electrical position:

$$\mathcal{L} = \begin{bmatrix} L_0 + L_2 \cos 2\theta & L_2 \sin 2\theta \\ L_2 \sin 2\theta & L_0 - L_2 \cos 2\theta \end{bmatrix} \quad (4)$$

where L_0 and L_2 represents the average and differential spatial inductances. Torque producing by the machine T_m is given as:

$$\begin{aligned}T_m &= \frac{3p}{2} [\psi_r (i_\beta \cos\theta - i_\alpha \sin\theta) \\ &\quad - L_2 ((i_\alpha^2 - i_\beta^2) \sin 2\theta - 2i_\alpha i_\beta \cos 2\theta)]\end{aligned}\quad (5)$$

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For a non-salient rotor PMSM, the equation (4) (5) can be simplified by:

$$L_2 = 0 \quad (6)$$

By taking:

$$x = [\mathcal{I}^T \quad \omega \quad \theta]^T ; \quad y = \mathcal{I} ; \quad u = \mathcal{V} \quad (7)$$

$$f(x, u) = \begin{bmatrix} \frac{d\mathcal{I}^T}{dt} & \frac{d\omega}{dt} & \frac{d\theta}{dt} \end{bmatrix} ; \quad h(x) = \mathcal{I} \quad (8)$$

system of the following form (denoted Σ) are discussed:

$$\Sigma : \begin{cases} \dot{x} = f(x(t), u(t)) \\ y = h(x(t)) \end{cases} \quad (9)$$

where $x \in X \subset \mathbb{R}^n$ is the state vector, $u \in U \subset \mathbb{R}^m$ is the input vector, $y \in \mathbb{R}^p$ is the output vector, f and h are C^∞ functions. The system of (9) can be applied to the PMSM model of (1).

In this report, a surface-mounted PMSM (SPMSM) is studied. The current state functions of which in stator reference frame are:

$$\begin{cases} \frac{di_\alpha}{dt} = -\frac{R_s}{L_0} i_\alpha + \frac{\psi_r}{L_0} \omega \sin\theta + \frac{u_\alpha}{L_0} \\ \frac{di_\beta}{dt} = -\frac{R_s}{L_0} i_\beta - \frac{\psi_r}{L_0} \omega \cos\theta + \frac{u_\beta}{L_0} \end{cases} \quad (10)$$

In order to design SPMSM observer, the discrete extended Kalman filter is applied. When sample period is small enough, ω is considered unchangeable, namely $\dot{\omega} = 0$.

3 PMSM Observability

The system which is shown in equation (1) is a 4th order system. Its observability matrix should contain the gradient of the output and its derivatives up to the 3th order. The partial observability matrix is [8] :

$$\mathcal{O}_{y1} = \frac{\partial(y, \dot{y})}{\partial x} = \begin{bmatrix} \mathbb{I}_2 & \mathbb{O}_{2 \times 1} & \mathbb{O}_{2 \times 1} \\ \frac{\partial}{\partial \mathcal{I}}(\frac{d\mathcal{I}}{dt}) & \frac{\partial}{\partial \omega}(\frac{d\mathcal{I}}{dt}) & \frac{\partial}{\partial \theta}(\frac{d\mathcal{I}}{dt}) \end{bmatrix} \quad (11)$$

where $\mathbb{I}_n$ is the $n \times n$ identity matrix, and $\mathbb{O}_{n \times m}$ is an $n \times m$ zero matrix, and:

$$\begin{aligned} \frac{\partial}{\partial \mathcal{I}}(\frac{d\mathcal{I}}{dt}) &= -\mathcal{L}^{-1} \mathcal{R}^{eq} \\ \frac{\partial}{\partial \omega}(\frac{d\mathcal{I}}{dt}) &= -\mathcal{L}^{-1} (\mathcal{L}' \mathcal{I} + \psi_r \mathcal{C}'(\theta)) \\ \frac{\partial}{\partial \theta}(\frac{d\mathcal{I}}{dt}) &= (\mathcal{L}^{-1})' \mathcal{L} \frac{d\mathcal{I}}{dt} - \mathcal{L}^{-1} (\mathcal{L}'' \mathcal{I} + \psi_r \mathcal{C}(\theta)) \omega \end{aligned} \quad (12)$$

where $\mathcal{L}'$ and $\mathcal{L}''$ denote the first and second partial derivatives of $\mathcal{L}$ with respect to θ :

$$\mathcal{L}' = \frac{\partial}{\partial \theta} \mathcal{L} ; \quad \mathcal{L}'' = \frac{\partial}{\partial \theta} \mathcal{L}' \quad (13)$$

The determinant Δ_{y1} of the sub-matrix which is shown in equation (11) is given by:

$$\begin{aligned} \Delta_{y1} &= \frac{1}{L_d L_q} [(L_\delta i_d + \psi_r)^2 + L_\delta^2 i_q^2] \omega \\ &+ \frac{L_\delta}{L_d L_q} \left[L_\delta \frac{di_d}{dt} i_q - (L_\delta i_d + \psi_r) \frac{di_q}{dt} \right] \end{aligned} \quad (14)$$

The observability condition $\Delta_{y1} \neq 0$ can be written as:

$$\omega \neq \frac{(L_\delta i_d + \psi_r) L_\delta \frac{di_q}{dt} - L_\delta \frac{di_d}{dt} L_\delta i_q}{(L_\delta i_d + \psi_r)^2 + L_\delta^2 i_q^2} \quad (15)$$

and it follows that:

$$\omega \neq \frac{d}{dt} \arctan \left(\frac{L_\delta i_q}{L_\delta i_d + \psi_r} \right) \quad (16)$$

The equation (16) defines a observability vector, denoted by $\Psi_\mathcal{O}$, that has the following components in the $d - q$ reference frame:

$$\begin{aligned} \Psi_{\mathcal{O}d} &= L_\delta i_d + \psi_r \\ \Psi_{\mathcal{O}q} &= L_\delta i_q \end{aligned} \quad (17)$$

and hence

$$\omega \neq \frac{d}{dt} \theta_\mathcal{O} \quad (18)$$

where $\theta_\mathcal{O}$ is the phase of the vector $\Psi_\mathcal{O}$ in the rotating reference frame.

For the SPMSM model:

$$L_d = L_q = L_0 \implies L_\delta = L_d - L_q = 0 \quad (19)$$

thus the equation (14) becomes:

$$\Delta_{y1} = \omega \left(\frac{\psi_r}{L_0} \right)^2 \quad (20)$$

and the observability vector is equal to rotor permanent magnet flux vector, which is fixed in the $d - q$ reference frame. The fictitious observability vector is shown in Fig. 1. Thus it can be obtained that the observability problem arises only at standstill situation.

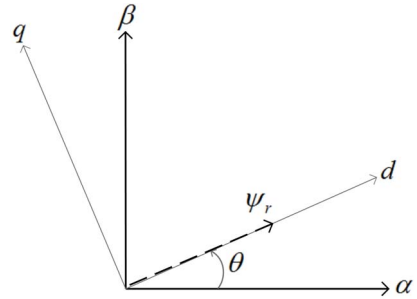


Fig. 1: Fictitious observability vector for SPMSM

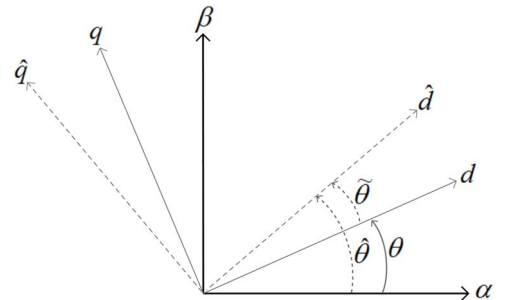


Fig. 2: Stator, estimated and real rotor reference frames for SPMSM

When it comes to the case of zero speed, state observer algorithms that combine high frequency injection (HFI) method were proposed. In [7] a high frequency voltage is injected into the motor and causes changes of current signal. Rotor position and speed including zero speed can be estimated based on natural or artificial motor salient effect.

To deal with the observability problem at zero speed, one solution combining HFI method with EKF algorithm is proposed [9]. A high frequency voltage is injected on the direct axis in the estimated rotating reference frame (Fig. 2), which results in a vibration of the rotor (nonzero speed) only if the position is not correctly estimated. If the following HF voltage:

$$v_d = V_{hf} \cos(\omega_{hf} t) \quad (21)$$

is injected in the estimated $\hat{d}\hat{q}$ rotating reference frame (obtained by the estimated position $\hat{\theta}$), the determinant Δ_{y1} becomes [8]:

$$\Delta_{y1} = -\frac{\psi_r^2}{L_0^2} \omega + \frac{\psi_r}{L_0^2} V_{hf} \cos(\omega_{hf} t) \sin \tilde{\theta} \quad (22)$$

where $\tilde{\theta}$ stands for the position estimation error. In the real Park reference frame, the injected signal is:

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \begin{bmatrix} V_{hf} \cos(\omega_{hf} t) \cos(\tilde{\theta}) \\ V_{hf} \cos(\omega_{hf} t) \sin(\tilde{\theta}) \end{bmatrix} \quad (23)$$

When the position is correctly estimated ($\tilde{\theta} = 0$), the d-axis injection has no effect on the movement of the machine. However, when the position is wrongly estimated, a high-frequency voltage component is generated on the real q-axis. This voltage produces a component of the magnetic field which is perpendicular to the rotor and therefore creates a torque that itself creates a vibration at the injection frequency. Thus the local observability is guaranteed at standstill.

4 EKF - based Observer Design

The Extended Kalman Filter is a set of mathematical equations which produce the optimal estimation of the state system based on least square method. The EKF estimates the process by using a feedback control. The EKF provides considerable good tolerance for the mathematical model errors and noises in the measurement inaccuracy [10].

The state space model of the simplified PMSM given by (9) is a nonlinear system. To apply the EKF algorithm, the system needs be discretized and linearized [11].

Based on equations (7) (8) (9), the state function of this nonlinear system is

$$\dot{x} = f(x) + Bu + v \quad (24)$$

where

$$f(x) = \begin{bmatrix} -\frac{R_s}{L_0} i_\alpha + \frac{\psi_r}{L_0} \omega \sin \theta \\ -\frac{R_s}{L_0} i_\beta - \frac{\psi_r}{L_0} \omega \cos \theta \\ 0 \\ \omega \end{bmatrix} \quad (25)$$

and

$$B = \begin{bmatrix} \frac{1}{L_0} & 0 \\ 0 & \frac{1}{L_0} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (26)$$

The output function is

$$y = h(x) + v \quad (27)$$

Here w and v are random disturbances. w is the process noise which stands for the errors of the parameters; v is the measurement noise which stands for the errors in the measurement and sample. The noise covariance matrixes are defined as follows:

$$\begin{aligned} Q &= \text{cov}(w) = E\{ww^T\} \\ R &= \text{cov}(v) = E\{vv^T\} \end{aligned} \quad (28)$$

where $E\{\cdot\}$ is the computation of expectation value.

From equation (24) (27), the discrete approximated equation is given by:

$$\begin{aligned} x_k &= (I + T_s f(x_{k-1}))x_{k-1} + T_s Bu_{k-1} + w_{k-1} \\ y_k &= h(x_{k-1})x_{k-1} + v_{k-1} \end{aligned} \quad (29)$$

After linearization of $f(x)$ and $h(x)$, the Jacobian matrix are obtained as follows:

$$F_k = \left. \frac{df}{dx} \right|_{x_k=\hat{x}_k} = \begin{bmatrix} -\frac{R_s}{L_0} & 0 & \frac{\psi_r}{L_0} \sin \theta & \omega \frac{\psi_r}{L_0} \cos \theta \\ 0 & -\frac{R_s}{L_0} & -\frac{\psi_r}{L_0} \cos \theta & \omega \frac{\psi_r}{L_0} \sin \theta \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (30)$$

$$H_k = \left. \frac{dh}{dx} \right|_{x_k=\hat{x}_k} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (31)$$

Define matrix P as the error covariance of observation

$$P_k = E\{e_k e_k^T\} = \sum_{i=1}^n E\{[x_i - \hat{x}_i][x_i - \hat{x}_i]^T\} \quad (32)$$

As the SPMSM model observer, the extended Kalman filter can be realized by iteration as follows:

a) Compute the state ahead and the error covariance ahead.

$$\begin{aligned} \hat{x}_{k/k-1} &= \hat{x}_{k-1} + T_s(f(x_{k-1}) + Bu_{k-1}) \\ P_{k/k-1} &= \Phi_{k/k-1} P_{k-1} \Phi_{k/k-1}^T + Q_{k-1} \end{aligned} \quad (33)$$

where Φ is transfer matrix from x_{k-1} to x_k , which is defined as

$$\Phi_{k/k-1} \approx I + F_{k-1} T_s \quad (34)$$

b) Compute the Kalman gain.

$$K_k = P_{k/k-1} H_k^T (H_k P_{k/k-1} H_k^T + R)^{-1} \quad (35)$$

c) Update estimation with measurement and error covariance matrix.

$$\begin{aligned} \hat{x}_k &= \hat{x}_{k/k-1} + K_k(y_k - H_k \hat{x}_{k/k-1}) \\ P_k &= P_{k/k-1} - K_k H_k P_{k/k-1} \end{aligned} \quad (36)$$

In the EKF observer design, choosing the proper values for the covariance matrices Q and R is of great importance. The change of values of covariance matrices affect both dynamic and steady state. By using trial and error methods, a suitable set of values of Q and R are selected to insure optimal stability and convergence time. In this work, the following matrices are used:

$$Q = \begin{bmatrix} 0.1 & 0 & 0 & 0 \\ 0 & 0.1 & 0 & 0 \\ 0 & 0 & 20 & 0 \\ 0 & 0 & 0 & 0.3 \end{bmatrix} ; \quad R = \begin{bmatrix} 0.2 & 0 \\ 0 & 0.2 \end{bmatrix} \quad (37)$$

5 Simulation Results

The PMSM model is constructed to verify the using conventional observer combining with EKF. The simulation is implemented using MATLAB/Simulink. The parameter of the motor model is given in Table 1. Based on optimal control theory, the desired control gains are determined .

Table 1: PMSM parameters

Variable	Physical Meaning	Value	Unit
R_s	Armature Resistance	2.125	Ω
L_0	Armature Inductance	$11.6e-3$	H
ψ_r	Rotor Magnetic Flux	0.387	$V/rad/s$
J	Moment of Inertia	0.03	$kg.m^2$
B	Damping Coefficient	0.121	$N.m/rad/s$
p	Poles Number	6	

The block diagram used in the simulation is as shown in Fig. 3. It is mainly composed of space vector pulse width modulation (SVPWM) block, calculation block, PI controller block, injected voltage block and EKF observer block.

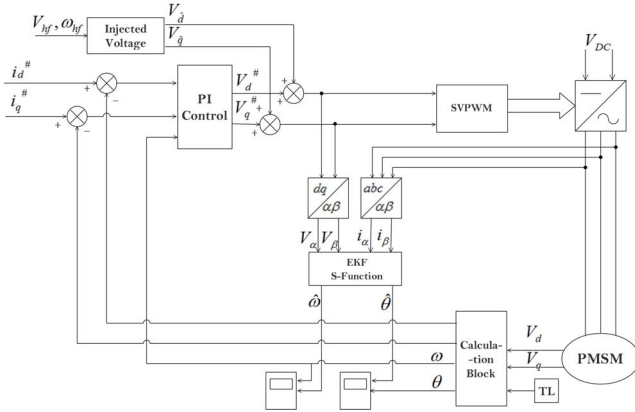


Fig. 3: Block diagram of PMSM with HFI and EKF

The EKF algorithm is implemented using S-Function block. The inputs of S-Function are reference voltages v_α and v_β under stator reference frame, and actual currents i_α and i_β taken directly from the machine terminals. The outputs of S-Function are estimated speed and position of the rotor $\hat{\omega}$ and $\hat{\theta}$, which are fed to injected voltage block in order to calculate the estimation error, meanwhile they are fed to scopes and are compared with real speed and position.

The PMSM system is controlled using PI controllers with the reference currents:

$$i_d^{\#} = 0A \quad ; \quad i_q^{\#} = 5A \quad (38)$$

Meanwhile v_d and v_q are obtained by PI controller, which are fed to SVPWM and calculation block, then control signals are obtained and are fed to inverter in order to realize the PMSM control. By applying the control method $i_d = 0$, the control of SVPWM, rotor speed and current are based on an estimator that is operated by PI controller. The structure of control system is shown as follows in Fig. 4, where L_q is the inductance of q-axis and ω_e is electrical speed.

The following HF voltage is injected in the estimated $\hat{d}\hat{q}$ rotating reference frame during the time period

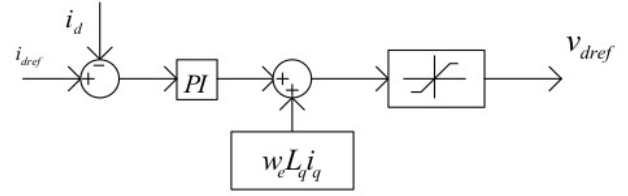


Fig. 4: Topology of PI controller for d-axis current

$[0.04s, 0.07s]$:

$$v_{\hat{d}} = 5\cos(2\pi.800t) \quad (39)$$

The Rotor observation results without HFI are shown in Fig. 5 – Fig. 8. The simulation results verify that the estimated rotor speed obtained by EKF is rather accurate with acceptable error and the estimated rotor position has obvious error at low speed.

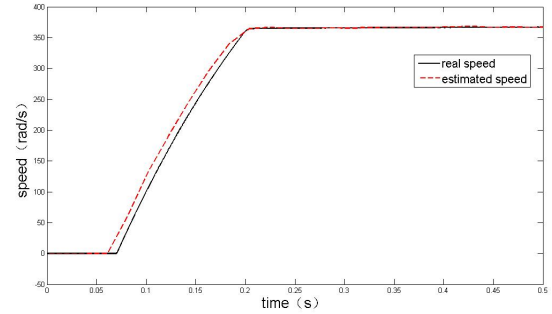


Fig. 5: Rotor real and estimated speed without HFI

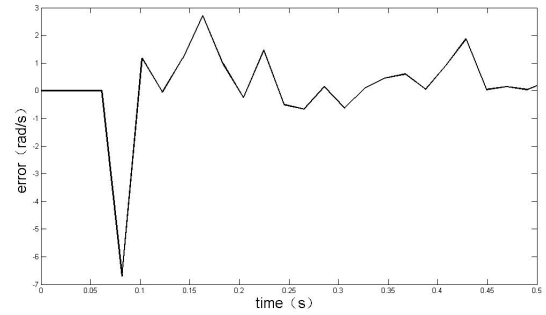


Fig. 6: Rotor speed estimation error without HFI

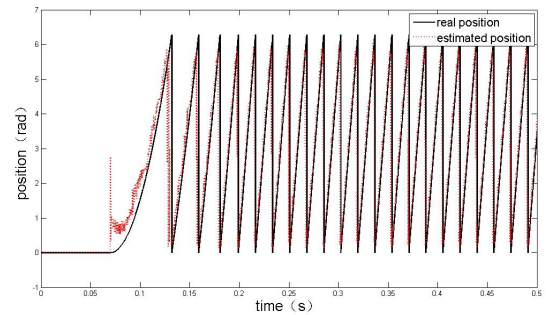


Fig. 7: Rotor real and estimated position without HFI

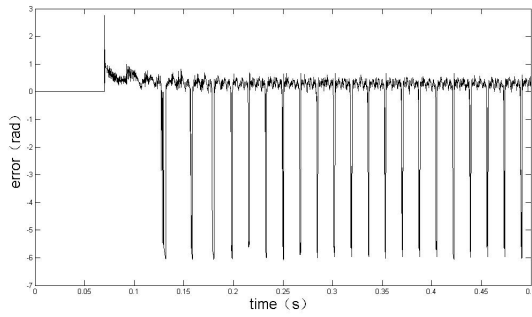


Fig. 8: Rotor position estimation error without HFI

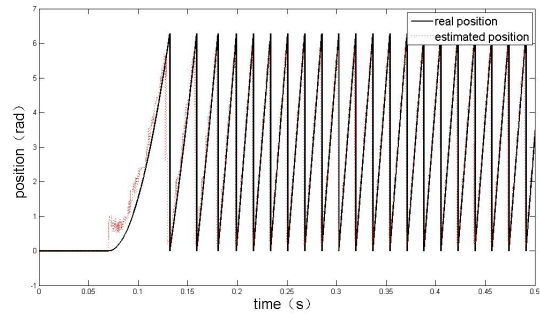


Fig. 11: Rotor real and estimated position with HFI

The Rotor speed observation results with HFI are shown in Fig. 9 – Fig. 12. It is obvious that after injecting HF voltage signal, the estimations of rotor position and speed when the SPMSM starts are more accurate and estimation errors are less both at standstill situation and in normal operating situation.

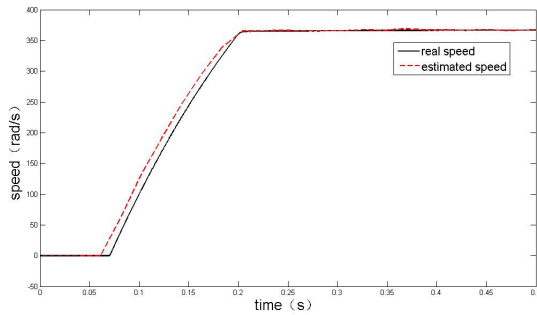


Fig. 9: Rotor real and estimated speed with HFI

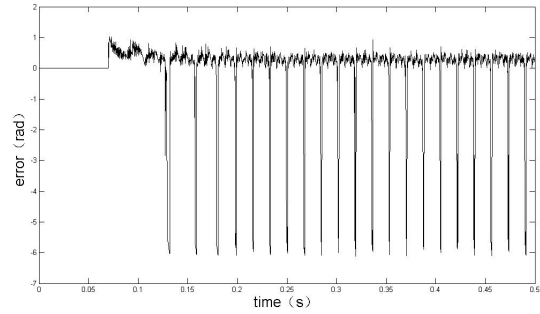


Fig. 12: Rotor position estimation error with HFI

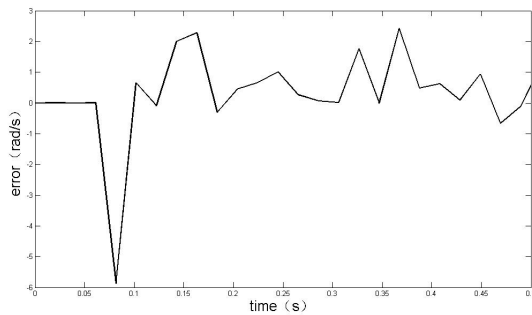


Fig. 10: Rotor speed estimation error with HFI

6 Conclusion

Using the presented approach, which applies sensorless control and observability analysis of permanent magnet synchronous machine based on extended Kalman filter and high frequency injection, it is possible to estimate the rotor speed and rotor position while the initial position and mechanic parameters are not needed. Furthermore, by injecting HF voltage, the estimation of extended Kalman filter is more accurate when SPMSM starts up. Though the problem is that the covariance matrixes of noises can only be determined by experiment since the noises and disturbances are not known in practice. The simulation result verifies the performance and the accuracy of the represented approach in dynamic state and zero speed.

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